Course: Mathematics of Music	CIE-2 - SCHEME	Maximum Marks: 10+50
Course Code: MA375TGR	Seventh Semester 2025-26 UG	Time: 1:30 PM to 3:30 PM Date: 16/12/2025

Q.No.	Questions	M
1.	Chakra 9 (Brahma), Position 4. Ma_2, Ri_3Ga_3, Dha_2Ni_2. Scale: Sa Ri_1 Ga_3 Ma_2 Pa Dha_2 Ni_2 Sa	2
2.	$\log_2(3/2) \approx 0.58496$. $a_0=0$, $a_1=1$, $a_2=1$. Answer: [0; 1, 1]	2
3.	$f(t) = a_0/2 + \Sigma[a_n\cos(2\pi nt/T) + b_n\sin(2\pi nt/T)]$ a_0: DC component, a_n: cosine amplitude, b_n: sine amplitude	2
4.	Definition: $SC = \Sigma(f_i \times A_i)/\Sigma(A_i)$ Calculation: $(5f+6f+3f)/(5+3+1) = 14f/9 \approx 1.556f$	2
5.	Pitch class: set of octave equivalents (0-11 for C-B). Addition: $p \oplus q = (p + q) \bmod 12$	2

Q.No.	Questions	M
1a	Proof: Assume $\log_2(3/2) = p/q$ rational. Then 2^(p+q) = 3^q. LHS even, RHS odd → Contradiction. ∴ irrational. Musical: $(3/2)^{12} \neq 2^7$ → Circle of fifths forms spiral. Pythagorean comma = 23.46 cents. No tuning has all perfect fifths AND octaves.	5 5
2a	Fixed: Sa, Pa. Variables: Ri(3), Ga(3), Ma(2), Dha(3), Ni(3). Constraint: Ri≤Ga, Dha≤Ni → 6 combinations each. Total: 2 × 6 × 6 = 72 Melakarthas. 12 chakras × 6 rāgas. Katapayadi: Consonant-to-digit mapping (ka→1, ta→6, etc.). Read in reverse order. Example: Dhīra-saṅ... → dha(4)ra(2) → 29.	6 4
3	Derivation: $y = bx^2$. Input: $x = A_1\cos(2\pi f_1t) + A_2\cos(2\pi f_2t)$. Apply $\cos^2(\theta) = ½[1+\cos(2\theta)]$ and $\cos(\alpha)\cos(\beta) = ½[\cos(\alpha-\beta)+\cos(\alpha+\beta)]$. Result: Difference tone at $\|f_1-f_2\|$, summation at f_1+f_2. Calculations: f_d = |800-600| = 200 Hz (G3). f_s = 800+600 = 1400 Hz (F6). Interval: 4/3 = Perfect Fourth. Organs/Brass: High SPL drives nonlinearity. Rich harmonics create many interaction products. Combination tones $\propto A_1A_2$.	4 3 3

4	Square wave odd function → only sine terms. $b_n = (4k/\pi n)$ for odd n, $b_n = 0$ for even n. $f(t) = (4k/\pi)\Sigma[\sin(2\pi(2m-1)f_0 t)/(2m-1)]$ Only odd harmonics (1,3,5,7,...) with amplitudes $\propto 1/n$. Sawtooth has ALL harmonics (1,2,3,4,...). Square: hollow, clarinet-like. Sawtooth: bright, string-like. Sawtooth perceived as twice as bright (denser spectrum).	6 4
5	$T_n(p) = (p+n) \bmod 12$. (i) Closure: $T_m \circ T_n = T_{(m+n)} \bmod 12$. (ii) Associativity: $((k+m)+n) = (k+(m+n))$ from integer addition. (iii) Identity: $T_0 \circ T_n = T_n \circ T_0 = T_n$. (iv) Inverse: $T_n \circ T_{(12-n)} = T_0$. Example: $T_5^{-1} = T_7$. Cyclic: Generator T_1 produces all elements ($T_1^n = T_n$). Isomorphism: $\varphi(n) = T_n$ proves $\{T_0,...,T_{11}\} \cong (\mathbb{Z}_{12},+)$.	2 2 2 2 2